
Outlier Migration in Long Decode Traces

Anonymous authors
Paper under double-blind review

Abstract

This draft reports the first two preregistered gates for OutlierMigrate, a candidate positive-method branch studying whether high-magnitude activation channels remain stationary during long decode. Phase 0 passes on Granite-4.0-H-Tiny: the fraction of top-1% position-100 channels whose rank changes by more than two channel positions by decode position 10000 is 0.8178385416666667 with bootstrap 95% CI [0.797265625, 0.8368489583333334]. Phase 1 replicates the dynamic-outlier direction at larger scale on Granite-4.0-H-Small: the migration fraction is 0.843165650406504 with bootstrap 95% CI [0.8334349593495936, 0.8511432926829268]. This is a strong positive-method finding for the same model family, but it is not camera-ready and does not yet support cross-model claims. Cross-model validation and audit checks remain required before submission-level claims.

1 Introduction

OutlierMigrate asks whether the channels carrying high-magnitude decode-time activations are fixed or whether they move during long reasoning traces. Static outliers would support static channel-protection recipes. Dynamic outliers would motivate migration-aware protection or routing methods, but only after validation shows that the effect is stable, useful, and not specific to one model family.

The current paper status is deliberately limited. Phase 0 is a screen on ibm-granite/granite-4.0-h-tiny with 12 deterministic AIME-2025 reasoning traces. Phase 1 scales the same dynamic path to ibm-granite/granite-4.0-h-small with 24 deterministic traces and an extended decode-position grid. Together they support a replicated same-family dynamic-outlier observation under the preregistered metric. They do not support a camera-ready method claim, an end-to-end efficiency claim, or any cross-model claim.

2 Related Work

Recent efficient-recurrence and linear-attention architectures make data-dependent channel-wise memory control a central design axis. Gated Linear Attention (GLA) formulates linear attention as recurrent matrix-valued state with data-dependent gating (Yang et al., 2023). Gated DeltaNet combines adaptive memory erasure with delta-rule updates and reports hybrids that mix Gated DeltaNet with attention or Mamba2 layers (Yang et al., 2024). Kimi Linear extends this direction: its Kimi Delta Attention (KDA) is described as an extension of Gated DeltaNet with finer-grained gating, and the released Kimi Linear model is a layerwise hybrid of KDA and Multi-Head Latent Attention (Zhang et al., 2025). The official Qwen3.6 model card likewise describes a hybrid layout that interleaves Gated DeltaNet blocks with gated attention (Qwen Team, 2026). These sources motivate OutlierMigrate as an empirical test of whether high-magnitude channels stay fixed during long decode in hybrid sequence models. They do not imply that Kimi Linear, Qwen3.6, GLA, or other unmeasured models exhibit the migration measured in this draft.

3 Method

Phase 0 follows the frozen preregistration at `experimental/outlier_migrate/phase0/preregister_om_phase0.md`. The model is `ibm-granite/granite-4.0-h-tiny` at HuggingFace snapshot commit `791e0d3d28c86e106c9b6e0b4cecddee0375b6124`. The prompt set contains 12 deterministically selected AIME-2025 traces, indices 0–11, with prompt SHA `sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd` from `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json`.

Phase 1 follows the frozen preregistration at `experimental/outlier_migrate/phase1/preregister_om_phase1.md`. The model is `ibm-granite/granite-4.0-h-small` at HuggingFace snapshot commit `b8c0982bab7fde4eb48110f5a069527c008fab39`. The prompt set contains 24 deterministically selected AIME-2025 traces, indices 0–23, with prompt SHA `sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e` from `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json`.

For every transformer layer, the runner records per-channel activation magnitudes. Phase 0 records decode positions {100, 500, 1000, 5000, 10000}; Phase 1 records {100, 500, 1000, 5000, 10000, 20000}. Each gate selects the top 1% channels by mean magnitude at position 100 and computes the fraction whose rank at the final preregistered position differs by more than two channel positions. The reported gate metric is the trace-bootstrap aggregate over layers with equal layer weighting.

4 Experiments

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. The checker output is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/artifact_check.json`. The activation artifact SHA is `sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687` from `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json`.

The Phase 1 result packet is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z`. The checker output is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/artifact_check.json`. The activation artifact SHA is `sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1` from `experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json`.

The preregistered dynamic-outlier pass rule in both gates requires migration fraction at least 0.05 and bootstrap CI lower bound greater than 0.05. Phase 1 uses the same dynamic path because Phase 0 returned `PASS_OM_PHASE0_DECODE_TIME_MIGRATION`.

5 Results

The exact Phase 0 provenance values are: model snapshot commit `791e0d3d28c86e106c9b6e0b4cecddee0375b6124`, prompt SHA `sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd`, and activation artifact SHA `sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687`; these are recorded in `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/model_provenance.json`, `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json`,

Quantity	Phase 0	Phase 1
Decision	dynamic-outlier pass	replicated-at-scale pass
Migration fraction	0.8178385416666667	0.843165650406504
Bootstrap 95% CI	[0.797265625, 0.8368489583333334]	[0.8334349593495936, 0.8511432926829268]
Preregistered threshold	≥ 0.05 and CI lower > 0.05	≥ 0.05 and CI lower > 0.05
Artifact complete	true	true
Model	ibm-granite/granite-4.0-h-tiny	ibm-granite/granite-4.0-h-small
Trace count	12	24
Layer count / hidden size	40 / 1536	40 / 4096
Decode positions	100, 500, 1000, 5000, 10000	100, 500, 1000, 5000, 10000, 20000
Bootstrap seed	20260508	20260508

Table 1: OutlierMigrate Phase 0 and Phase 1 gate results. Phase 0 values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json. Phase 1 values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json.

and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json. The exact Phase 1 provenance values are: model snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39, prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e, and activation artifact SHA sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1; these are recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/model_provenance.json, experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json, and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

The Phase 0 checker returned `PASS_OM_PHASE0_DECODE_TIME_MIGRATION`. Its exact artifact reason was that the point estimate 0.81783854 is at least 0.05 and the CI lower bound 0.79726562 is greater than 0.05. The Phase 1 checker returned `PASS_OM_PHASE1_REPLICATED_AT_SCALE`. Its exact artifact reason was that the point estimate 0.84316565 is at least 0.05 and the CI lower bound 0.83343496 is greater than 0.05.

6 Discussion

The Phase 0 result weakens the static-outlier hypothesis on the small Granite surface: under the preregistered rank-migration metric, most position-100 top-1% channels do not remain rank-stationary through position 10000. Phase 1 promotes the dynamic-outlier hypothesis within the Granite family because the effect direction replicated on a larger model, twice as many traces, and an extended decode-position grid.

This is the strongest OutlierMigrate evidence so far and keeps the branch alive as a positive-method candidate. It does not establish that migration-aware intervention improves model quality, latency, memory, or robustness. The highest-priority next gate is cross-model validation with audit checks that separate a reusable migration mechanism from a same-family artifact.

The theoretical mechanism is plausible but not yet empirically validated outside Granite. In channel-wise gated recurrence or linear-attention layers, independent per-channel decay can make low-decay channels accumulate magnitude over long decode while high-decay channels shed magnitude. Under that mechanism, the top 1% channels by magnitude at position 100 need not remain the top 1% at later positions. Kimi Linear’s KDA, Qwen3.6-style Gated

DeltaNet blocks, and GLA-style gated recurrence provide architectural examples where channel-wise or fine-grained data-dependent gating is a first-class operation (Zhang et al., 2025; Qwen Team, 2026; Yang et al., 2024; 2023). This paragraph is a mechanism argument only; this draft has not measured migration on Kimi Linear, Qwen3.6, or GLA variants.

7 Limitations

This draft is not camera-ready. It reports two positive gates in one model family. There is no cross-model validation, no strict cross-family falsification pair, no contamination audit, no independent seed repeat beyond the recorded bootstrap procedure, and no intervention showing that migration-aware handling improves an application metric. The result may also depend on the chosen decode positions, top-channel fraction, and rank-delta threshold, all of which were fixed before analysis and should not be retuned on these packets.

The empirical measurement here is on Granite Mamba-2-style SSM hybrids. The planned Phase 2 cross-validation should extend the same preregistered logic to delta-rule linear-attention hybrids such as Qwen3.6 and Kimi Linear before the paper claims architectural generality. Generalization to other gated recurrence variants, including RWKV-7 and GLA-style systems, is not established by these results.

8 Reproducibility Statement

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om__phase0_20260508T011824Z`. It includes `environment.json`, `model_provenance.json`, `prompt_manifest.json`, `command_metadata.json`, `random_seed.json`, `metrics.json`, `bootstrap_ci.json`, `checker_result.json`, and `artifact_check.json`. The recorded Phase 0 command was:

```
experimental/shared/run_phase0_branch.py --branch outlier_migrate
```

The Phase 1 result packet is `experimental/outlier_migrate/phase1/results/om__phase1_20260508T014959Z`. It includes the same artifact classes and records the Granite-4.0-H-Small snapshot commit `b8c0982bab7fde4eb48110f5a069527c008fab39`. The recorded Phase 1 command was:

```
experimental/outlier_migrate/phase1/run_om_phase1.py
```

The checker decisions and exact numbers in this draft are taken from the two result directories above.

9 Ethics Statement

This is an activation-analysis systems study on benchmark prompts. The current artifact does not make deployment, user-data, safety, or capability-improvement claims. The main ethical risk is overclaiming same-family Phase 0 and Phase 1 evidence as a validated cross-model positive method; this draft explicitly rejects that interpretation until cross-model validation and audits pass.

References

- Qwen Team. Qwen3.6-35B-A3B model card. Hugging Face, 2026. <https://huggingface.co/Qwen/Qwen3.6-35B-A3B>.
- Songlin Yang, Bailin Wang, Yikang Shen, Rameswar Panda, and Yoon Kim. Gated Linear Attention Transformers with Hardware-Efficient Training. arXiv:2312.06635, 2023. <https://arxiv.org/abs/2312.06635>.
- Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated Delta Networks: Improving Mamba2 with Delta Rule. arXiv:2412.06464, 2024. <https://arxiv.org/abs/2412.06464>.

Kimi Team: Yu Zhang et al. Kimi Linear: An Expressive, Efficient Attention Architecture.
arXiv:2510.26692, 2025. <https://arxiv.org/abs/2510.26692>.